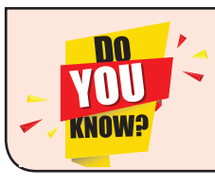
and capsule. They have more differentiated structure than liverworts. They often form dense cushions. Example: *Funaria*.

### 2.4.3 Economic importance

Dead thalli of *Sphagnum* gets accumulated and compressed, hardened to form peat. In northern Europe (Netherlands)peat is used as fuel in commercial scale. Apart from this nitrates, brown dye and tanning materials are derived from peat. *Sphagnum* and peat are also used in horticulture as packing material because of their water holding capacity. *Marchantia polymorpha* is used to cure pulmonary tuberculosis. *Sphagnum*, *Bryum* and *Polytrichum* are used as food. Bryophytes play a major role in soil formation through succession and help in soil conservation.




*Buxbaumia aphylla* and *Cryptothallus mirabilis* are saprophytic bryophytes

### 2.4.4 Marchantia

Class - Hepaticopsida
Order - Marchantiales
Family - Marchantiaceae
Genus - *Marchantia*

*Marchantia* grows in cool moist shady places. *Marchantia polymorpha* is the common species.

**Gametophyte**

The plant body of *Marchantia* is a gametophyte. It is prostrate,dorsiventral and dichotomously branched. The thallus on the dorsal surface possess conspicuous median midrib which is marked by a shallow groove on dorsal surface. The dorsal surface appears to have rhomboidal or polygonal diamond shaped areas which indicate the outline of the underlying air chambers of the thallus (Figure 2.9).

The dorsal surface also shows crescent shaped structures called gemma cups which contain vegetative reproductive structures called gemmae. The apical notch bears an apical cell which helps in the growth of the thallus.The ventral surface the thallus bears multicellular scales and rhizoids which help in fixation and absorption of water and minerals. The rhizoids are of two types namely smooth walled or pegged (tuberculate) type. On maturation the thallus bears erect antheriophores and archegoniophores.

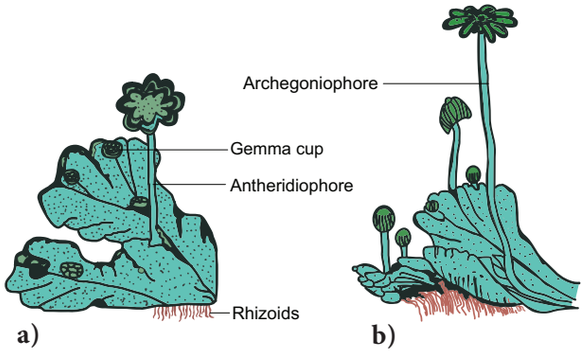



**Figure 2.9:** *Marchantia*
**a)** Thallus with antheridiophore
**b)** Thallus with archegoniophore

**Internal structure of Thallus**

In transverse section the *Marchantia* thallus shows three parts namely: epidermis, photosynthetic region and storage region (Figure 2.10).

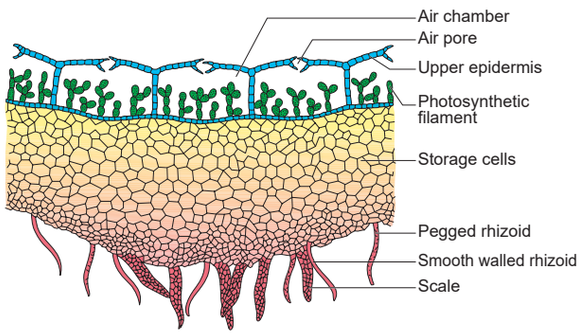



**Figure 2.10:** T.S. of Thallus

The epidermis has the upper and lower layers. The upper epidermis is single layered with thin walled parenchymatous cells. The cells possess chloroplasts. The upper epidermis is interrupted by many barrel shaped air pores which communicate with the air chambers. The pore is surrounded by 4 to 8 superimposed tiers of cells. Below the upper epidermis a number of air chambers are present in a single horizontal layer.



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